

REPORT ON
CMOS LOW NOISE AMPLIFIER ANALYSIS USING S- PARAMETER



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Department of Electronics and Communication Engineering

Neerukonda, Mangalagiri Mandal, Guntur District, Andhra Pradesh, 522502, India.

Submitted to:

Dr. Bharath Bhushan Upadhyay

Assistant Professor

Dept of ECE

Submitted by:

R.Tanish - AP23110020079

SK.Rehaan - AP23110020082

G.Sriram - AP23110020093

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1. Abstract

The rapid growth of wireless communication systems has increased the demand for efficient and low-noise RF front-end circuits. Among these, the Low Noise Amplifier (LNA) plays a crucial role as it is the first active block in a receiver chain, responsible for amplifying weak incoming signals while introducing minimal noise. In this project, a CMOS-based Low Noise Amplifier is designed and analyzed using S-parameter techniques.

The proposed LNA employs a common-source topology with source degeneration, which improves input impedance matching, enhances stability, and reduces noise figure. The circuit is designed to operate at a frequency of approximately 2.45 GHz, which lies in the Industrial, Scientific, and Medical (ISM) band commonly used for Wi-Fi, Bluetooth, and other wireless applications. CMOS technology is chosen due to its advantages such as low power consumption, low cost, and ease of integration with other digital and analog circuits.

The performance of the LNA is evaluated using S-parameters, which are widely used in RF circuit analysis. Key parameters such as S_{11} (input reflection coefficient), S_{21} (forward gain), S_{12} (reverse isolation), and S_{22} (output reflection coefficient) are analyzed over a frequency range of 0 to 5 GHz. The results indicate that the designed LNA achieves good input matching with S_{11} less than -10 dB at the desired operating frequency, ensuring efficient power transfer.

The forward gain (S_{21}) demonstrates moderate amplification suitable for RF applications, while the reverse isolation (S_{12}) is significantly low, indicating minimal feedback and improved stability. Although the output matching (S_{22}) is acceptable, further optimization can enhance overall performance. The inclusion of inductors and capacitors in the design enables proper impedance matching and frequency tuning.

Overall, the designed CMOS LNA provides a balanced trade-off between gain, stability, and matching performance. The results confirm that the proposed design is suitable for RF front-end applications in modern wireless communication systems. Future improvements can focus on increasing gain, reducing noise figure, and optimizing output matching for enhanced efficiency.

2. Introduction

The advancement of wireless communication systems has created a strong demand for high-performance RF front-end circuits. One of the most important components in a receiver is the Low Noise Amplifier (LNA), which is responsible for amplifying weak incoming signals without significantly degrading the signal quality. Since the LNA is the first stage in the receiver chain, its performance directly affects the overall sensitivity and noise performance of the system. Therefore, designing an efficient LNA with low noise, good gain, and proper impedance matching is essential for reliable communication.

With the continuous scaling of semiconductor technology, CMOS has become a preferred choice for RF circuit design due to its advantages such as low cost, low power consumption, and high integration capability. In this project, a CMOS-based LNA is designed using a common-source topology with source degeneration to improve stability and input matching. The performance of the amplifier is evaluated using S-parameters, which provide insight into reflection, gain, and isolation characteristics. The circuit is designed to operate around 2.45 GHz, making it suitable for applications such as Wi-Fi and Bluetooth communication systems.

3. Objectives

The main objective of this project is to design and analyze a CMOS-based Low Noise Amplifier (LNA) suitable for RF front-end applications. The LNA is intended to amplify weak incoming signals with minimal addition of noise, thereby improving the overall sensitivity and performance of the receiver system. Special emphasis is given to achieving efficient operation at a frequency of approximately 2.45 GHz, which is widely used in wireless communication systems such as Wi-Fi and Bluetooth.

Another important objective is to ensure proper impedance matching at both the input and output of the amplifier to maximize power transfer and minimize signal reflections. This is achieved using matching networks and source degeneration techniques. The project also aims to obtain sufficient gain while maintaining stability and low noise performance. The use of CMOS technology further focuses on reducing power consumption and enabling easy integration with other circuits.

Additionally, the project involves analyzing the performance of the LNA using S-parameters, including S_{11} , S_{21} , S_{12} , and S_{22} , over a specified frequency range. These parameters help in evaluating input matching, gain, reverse isolation, and output matching of the circuit. The overall goal is to design a stable, efficient, and practical LNA that meets the requirements of modern RF communication systems.

4. Theory

The CMOS Low Noise Amplifier (LNA) circuit consists of various components such as MOSFETs, inductors, capacitors, resistors, and power supply. Each component plays an important role in amplification, impedance matching, and stability of the circuit.

The **NMOS transistor (M1)** is the main active element of the circuit. It operates as a common-source amplifier and is responsible for amplifying the weak input RF signal. When the input signal is applied to the gate terminal, it controls the current flowing from drain to source, thereby producing an amplified output signal.

The **second NMOS transistor (M2)** acts as a current source or biasing element. It helps in maintaining a constant current in the circuit and ensures that the main transistor operates in the correct region (saturation region) for proper amplification.

The **input capacitor (C1)** is used as a coupling capacitor. It allows the AC signal to pass into the circuit while blocking the DC component, thus protecting the biasing of the transistor. Similarly, the **output capacitor (C4)** is used to couple the amplified signal to the output port without disturbing the DC conditions of the circuit.

The **bypass capacitor (C3)** is used to provide AC grounding. It reduces noise and stabilizes the circuit by filtering unwanted signals in the biasing network.

The **inductors (L1, L2, L3)** play a key role in RF circuit design. The **drain inductor (L1)** acts as a load and helps in increasing the voltage gain of the amplifier. The **input inductor (L2)** is used for impedance matching between the source and the amplifier input. The **source inductor (L3)** provides source degeneration, which improves input matching, reduces noise, and enhances stability.

The **resistor (R0)** is used for biasing the transistor. It helps in setting the correct operating point by controlling the gate voltage.

The **DC voltage source (VDD)** supplies power to the circuit and enables the transistor to operate.

The **input port (PORT0)** is where the RF signal is applied, and the **output port (PORT1)** is where the amplified signal is obtained. These ports are usually matched to 50 ohms to ensure maximum power transfer.

Overall, each component in the CMOS LNA circuit contributes to achieving proper amplification, low noise performance, good impedance matching, and stable operation at high frequencies.

5. Circuit Description

The given circuit is a CMOS Low Noise Amplifier (LNA) designed using a common-source NMOS configuration with source degeneration. The input RF signal is applied at the gate of the main transistor through a coupling capacitor, which blocks DC and allows only the AC signal to pass. The transistor amplifies the signal, and the output is taken from the drain through another coupling capacitor. Inductors are used for impedance matching and tuning the circuit to the desired frequency, while the source inductor improves stability and reduces noise. A second transistor and resistor provide proper biasing to ensure correct operation. The circuit operates with a DC supply and is designed to achieve good gain, input matching, and low noise performance at RF frequencies.

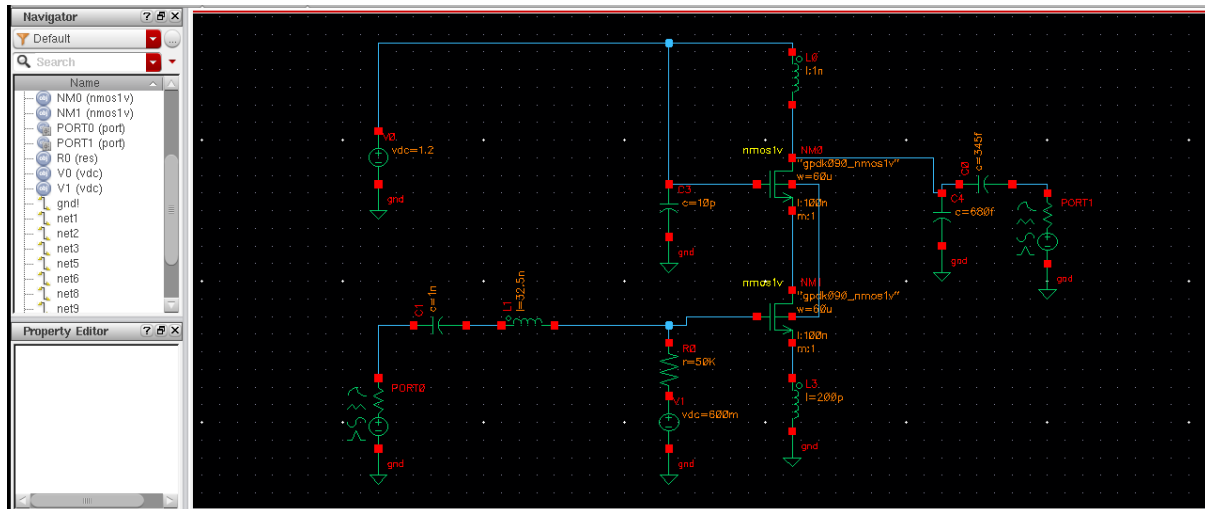
6. Working

When the RF input signal is applied to the circuit through the input port, it passes through the coupling capacitor and reaches the gate of the NMOS transistor. The transistor amplifies this weak signal by controlling the current flow from drain to source based on the input voltage. The source inductor provides stabilization and improves input matching, while the drain inductor acts as a load to enhance the voltage gain. The amplified signal is then taken from the drain and passed through the output coupling capacitor to the output port. Proper biasing ensures that the transistor operates in the correct region, resulting in efficient amplification with low noise.

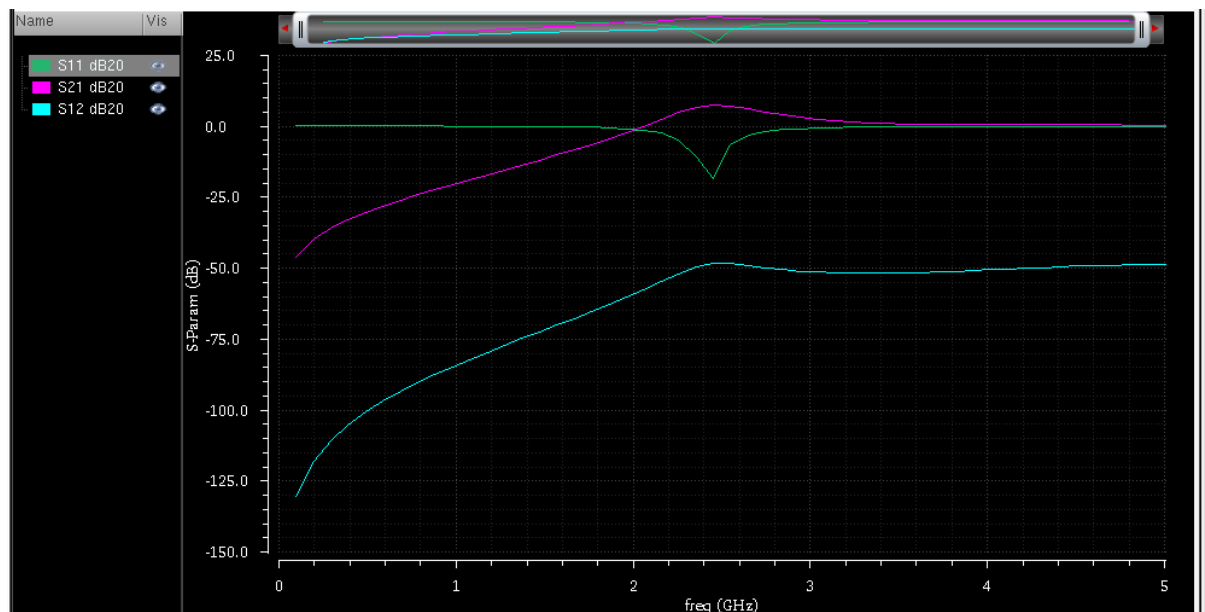
7. Simulation

The CMOS LNA circuit was simulated using RF simulation software to analyze its performance over a frequency range of 0 to 5 GHz. Proper DC biasing was applied to ensure correct operation of the transistor. S-parameter analysis was performed to evaluate important characteristics such as input matching (S11), gain (S21), reverse isolation (S12), and output matching (S22). The results were observed through graphs, and the circuit showed good input matching, moderate gain, and excellent isolation at the operating frequency of around 2.45 GHz. The simulation confirms that the designed LNA performs effectively for RF applications.

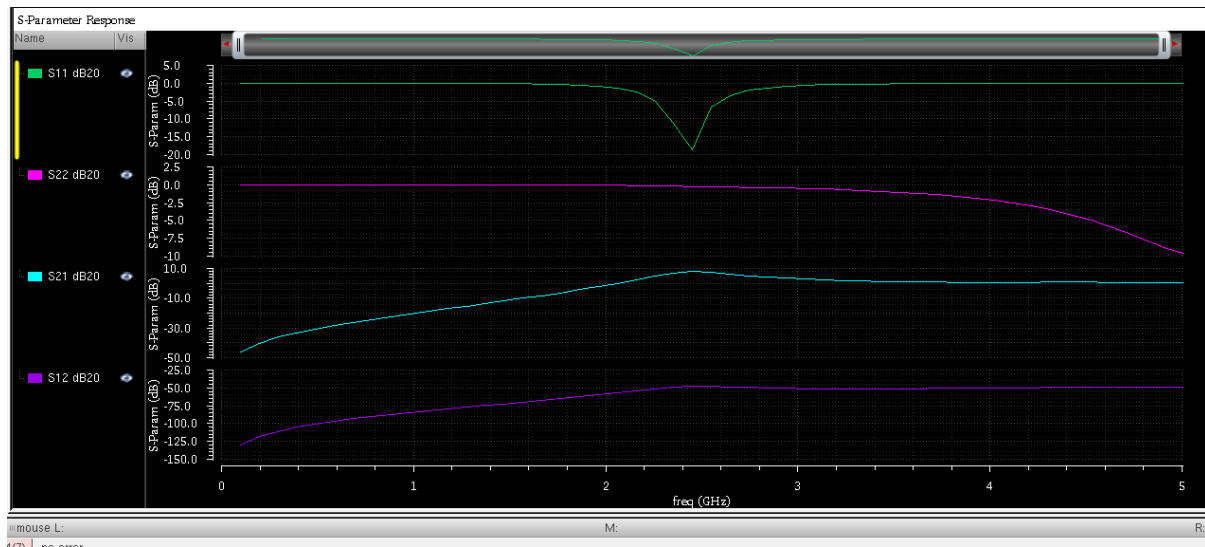
8. Results



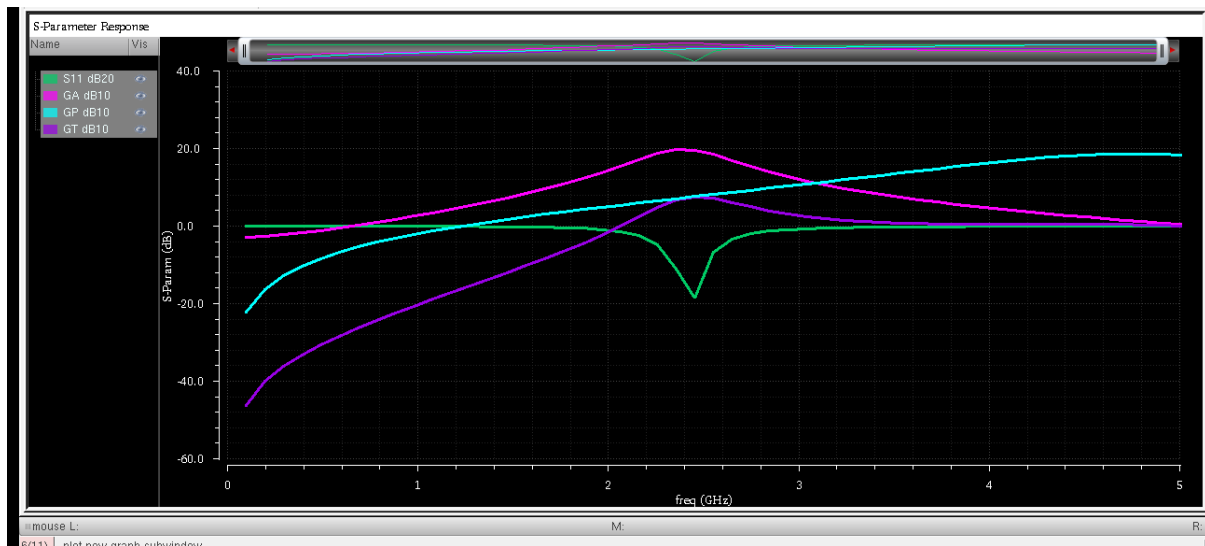
(A) This shows the CMOS LNA circuit with MOSFET, inductors, capacitors, and biasing network used for simulation.



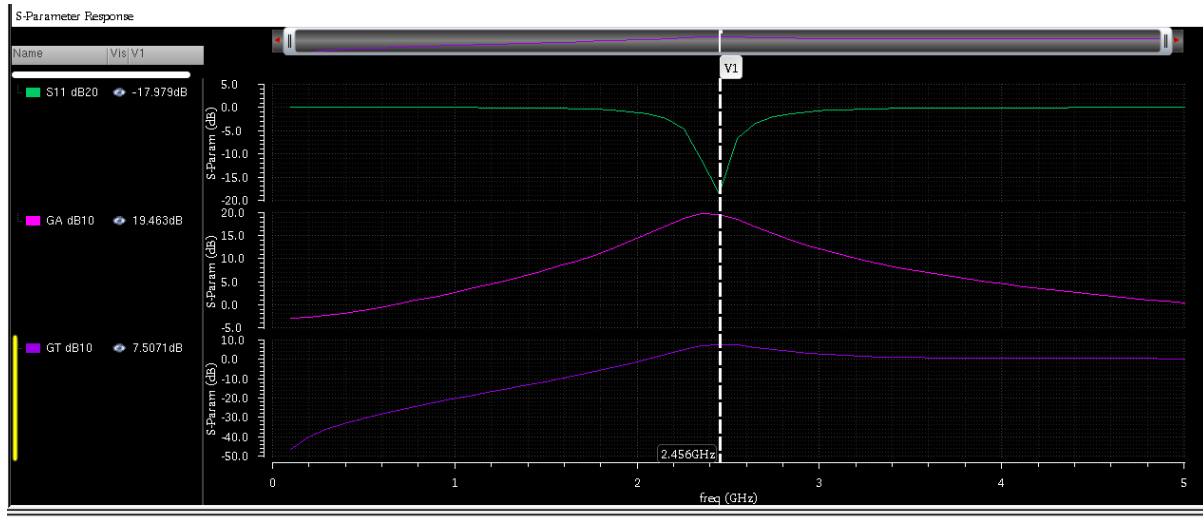
(B) This graph shows input matching, gain, and isolation of the LNA. S11 dips around 2.45 GHz indicating good matching, S21 shows amplification, and S12 is very low showing good isolation.



(C) This graph represents all S-parameters of the circuit. Input matching is good, gain is moderate, and output matching is acceptable with low reverse isolation.



(D) This graph shows different gain values of the LNA. Available gain is higher, while transducer gain is lower due to mismatch losses.



(E) This shows performance at operating frequency. S11 is around -18 dB and gain values confirm proper working of the amplifier.

9. Discussion

The simulation results show that the designed CMOS Low Noise Amplifier performs effectively at the target frequency of around 2.45 GHz. The input reflection coefficient (S11) is less than -10 dB, indicating good impedance matching and efficient power transfer. The forward gain (S21) demonstrates moderate amplification suitable for LNA applications, while the reverse isolation (S12) is very low, ensuring stability and minimal feedback. Although the output matching (S22) is acceptable, it can be further improved for better performance. Overall, the use of source degeneration and proper matching networks helps achieve a balanced performance in terms of gain, stability, and matching, making the circuit suitable for RF front-end applications.

10. Conclusion

The designed CMOS Low Noise Amplifier successfully operates at around 2.45 GHz and demonstrates good performance in terms of input matching, gain, and stability. The S-parameter analysis shows that the circuit achieves efficient power transfer with S11 less than -10 dB, moderate gain, and excellent reverse isolation. The use of source degeneration improves stability and noise performance, making the design reliable for RF applications. Although there is scope for improving output matching and gain, the overall results indicate that the LNA is suitable for wireless communication systems such as Wi-Fi and Bluetooth.

11. Advantages

- Low noise performance improves signal quality
- Good input impedance matching ($S_{11} < -10$ dB)
- Provides moderate and stable gain
- Excellent reverse isolation (very low S_{12})
- High stability due to source degeneration
- Low power consumption using CMOS technology
- Simple and compact circuit design
- Suitable for RF applications like Wi-Fi and Bluetooth
- Easy integration with other circuits
- Cost-effective implementation

12. Limitations

- Moderate gain (not very high for some applications)
- Output matching (S_{22}) is not fully optimized
- Performance depends strongly on component values
- Sensitive to process variations in CMOS technology
- Limited bandwidth around the operating frequency
- Noise figure can still be improved further
- Inductors occupy more area in integrated circuits
- Requires careful tuning for desired frequency

13. Applications

- Wireless communication systems (Wi-Fi, Bluetooth)
- RF receiver front-end circuits
- Mobile communication devices
- Satellite communication systems
- Radar systems
- IoT (Internet of Things) devices
- GPS receivers
- Wireless sensor networks
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